Bao Khuong

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Education

University of South Florida — Tampa, FL

Bachelor of Science in Computer Science | ***GPA: 3.94*** — *Aug. 2023 – May 2027*

Projects

ADVERSARIAL PROMPT ATTACKS ON ALIGNED LLMS | *Python, transformers* — April 2026

- Comparing Attack Success Rate (ASR) of an abliterated vs. aligned attacker model (Llama-3-8B) against a target model (Gemma-2-9B) across iterative refinement loops to evaluate the impact of model censorship on attack effectiveness.
- Classifying successful adversarial prompts into a semantic strategy taxonomy (persona adoption, hypothetical framing, authority roleplay, obfuscation, indirect elicitation) to identify which strategies yield the highest ASR.
- Analyzing cross-model transferability of adversarial prompts across same-parameter-count architectures under black-box conditions.

MATHEMATICAL REASONING IN SMOLLM2-360M VIA GRPO |
Python, PyTorch, HuggingFace TRL — April 2026

- Built a 5-agent closed-loop agentic AI system (Observer, Planner, Simulator, Evaluator, Adapter) that analyzes productivity logs, generates optimization strategies, and runs what-if simulations to project outcomes before recommending them.
- Conducted a reward shaping study comparing GRPO variants against SFT baseline across 600 test samples, finding that SFT remains a hard-to-beat baseline at 360M scale and that reward complexity beyond R2 destabilizes the policy rather than improving it.
- Demonstrated that length-penalized reward (R4) reduced average reasoning chain length by 18% at the cost of lowest accuracy across all variants, suggesting conciseness and correctness are in direct tension at the 360M parameter scale.

TRANSFORMER LANGUAGE MODEL FROM SCRATCH | *Python, CuPy* — March 2026

- Implemented a full **GPT2-style transformer** from scratch using only CuPy (GPU-accelerated NumPy), including multi-head causal self-attention, pre-norm transformer blocks, and AdamW optimizer with warmup LR scheduling without any deep learning frameworks.
- Engineered all components manually including forward and backward passes for attention, GELU/dropout/layer norm, sinusoidal positional encoding, and cross-entropy loss with label smoothing, achieving a validation set perplexity of **8.82** on character-level Tiny Shakespeare.
- Trained a **11M** parameter model on an A100 GPU, applying techniques such as gradient clipping, weight decay, and inverted dropout to stabilize training and reduce overfitting.

MULTI-AGENT SYSTEM PRODUCTIVITY OPTIMIZATION | *Python, Gemini API, Streamlit* April 2026

- Built a **5-agent** closed-loop agentic AI system (Observer, Planner, Simulator, Evaluator, Adapter) that analyzes productivity logs, generates optimization strategies, and runs what-if simulations to project outcomes before recommending them.
- Designed a data-driven scoring system with dynamically updating evaluation weights based on predicted-vs-actual metric deltas, enabling recommendations to personalize over iterative cycles.
- Validated on a synthetically generated dataset with engineered inefficiencies, achieving 8% completion rate improvement on the top strategy; visualized full agent reasoning chain via Streamlit dashboard.

Technical Skills

Languages: Python, C/C++, Java, JavaScript, HTML/CSS
Libraries & Frameworks: PyTorch, TensorFlow, Keras, CuPy, NumPy, Transformers, Matplotlib, OpenCV, Eigen, Node.js, AWS, Microsoft Azure, Git, Linux